MATH 15200: Calculus II
Lecture Notes: Inverse Functions, Inverse Trigonometric Functions, and Integration by Parts

Salas, Hille, and Etgen, §§7.1, 7.7, and 8.2

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1 Inverse functions

Definition 1.1. A function f is *one-to-one* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$; that is, f never takes the same value at two different points. If f is one-to-one with domain D and range R, its *inverse* is the function $f^{-1}\colon R \to D$ defined by

$$f^{-1}(y) = \text{the unique } x \in D \text{ with } f(x) = y.$$

The following facts come straight from the definition.

- $f^{-1}\big(f(x)\big) = x$ for every $x \in D$, and $f\big(f^{-1}(y)\big) = y$ for every $y \in R$.
- The domain of f^{-1} is the range of f, and the range of f^{-1} is the domain of f.
- (a, b) lies on the graph of f if and only if (b, a) lies on the graph of f^{-1}. So the graph of f^{-1} is the reflection of the graph of f across the line $y = x$.
- Warning: $f^{-1}(x)$ does not mean $1/f(x)$.

Example 1.2. (a) $f(x) = 2x + 3$ is one-to-one on $\mathbb{R}$. To find f^{-1}, solve $y = 2x + 3$ for x to get $x = \frac{y-3}{2}$. So $f^{-1}(x) = \frac{x-3}{2}$.

(b) $f(x) = x^2$ is not one-to-one on $\mathbb{R}$, since $f(-1) = f(1)$. It is one-to-one on $[0, \infty)$, and the inverse of this restricted function is $\sqrt{x}$. We will do the same thing with the trigonometric functions in Section 3: restrict the domain to make the function one-to-one, then invert.

Monotonic functions are one-to-one.

Theorem 1.3. If f is strictly increasing or strictly decreasing on its domain, then f is one-to-one.

Proof. Let $x_1 \neq x_2$, say $x_1 < x_2$. Then either $f(x_1) < f(x_2)$ or $f(x_1) > f(x_2)$; in either case $f(x_1) \neq f(x_2)$. □

Recall from the Mean Value Theorem that if $f' > 0$ on an interval, then f is strictly increasing there, and if $f' < 0$, then f is strictly decreasing. This stays true if f' vanishes at finitely many points (apply the MVT on each piece). For instance, x^3 is strictly increasing on $\mathbb{R}$ even though its derivative is 0 at 0.

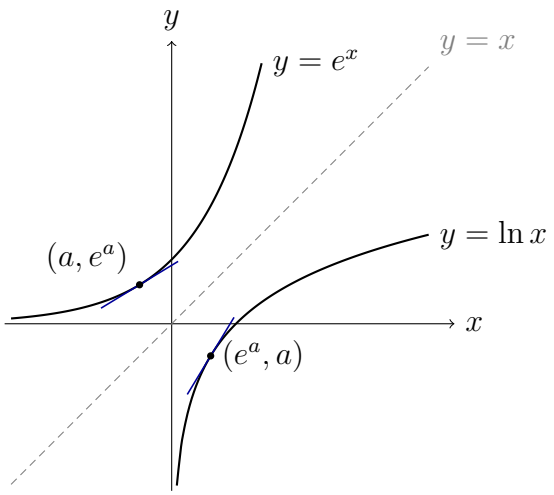
Remark. Conversely, a continuous function that is one-to-one on an interval must be strictly increasing or strictly decreasing there (this follows from the Intermediate Value Theorem). We won't need this: in practice, we check that a function is one-to-one by checking the sign of its derivative.

Example 1.4. Let $f(x) = x^3 + x$. Since $f'(x) = 3x^2 + 1 > 0$, f is strictly increasing, and so one-to-one, on $\mathbb{R}$. Its range is all of $\mathbb{R}$: f is continuous and $f(x) \to \pm\infty$ as $x \to \pm\infty$, so by the IVT it takes every real value. So f^{-1} is defined on all of $\mathbb{R}$.

However, solving $y = x^3 + x$ for x requires the cubic formula, so there is no convenient formula for f^{-1}. We can still read off some values: $f(0) = 0$, $f(1) = 2$, and $f(2) = 10$, so $f^{-1}(0) = 0$, $f^{-1}(2) = 1$, and $f^{-1}(10) = 2$.

Question: What is the slope of the graph of f^{-1} at $x = 2$? We will answer this in Example 2.4, without ever finding a formula for f^{-1}.

Exponential and logarithm. Recall that ln is strictly increasing on $(0, \infty)$, since its derivative is $1/x > 0$, that its range is $\mathbb{R}$, and that its inverse is the exponential function e^x. So $e^{\ln x} = x$ for $x > 0$ and $\ln(e^x) = x$ for all x, and the graphs of e^x and $\ln x$ are reflections of each other across $y = x$.



The tangent lines at (a, e^a) and (e^a, a) are also reflections of each other. Reflecting across $y = x$ swaps rise and run, so their slopes, e^a and $1/e^a$, are reciprocals. This is the idea behind the next section.

2 Derivatives of inverse functions

The formula, if we assume the derivative exists. Suppose f is one-to-one and differentiable, and suppose we already knew that f^{-1} is differentiable. Differentiating both sides of $f\big(f^{-1}(x)\big) = x$ with the chain rule gives

$$f'\big(f^{-1}(x)\big) \cdot (f^{-1})'(x) = 1, \qquad \text{so} \qquad (f^{-1})'(x) = \frac{1}{f'\big(f^{-1}(x)\big)}.$$

This is the correct formula, and this computation is the easiest way to remember it. But it is not a proof: it *assumes* that f^{-1} is differentiable, and that is the main thing to prove.

Theorem 2.1 (Derivative of an inverse)**.** Let f be differentiable and strictly monotonic on an open interval I, let $J = f(I)$, and let $g = f^{-1}\colon J \to I$. Let $a \in I$ and $b = f(a)$. If $f'(a) \neq 0$, then g is differentiable at b, and

$$g'(b) = \frac{1}{f'(a)} = \frac{1}{f'\big(g(b)\big)}.$$

The proof rests on the continuity of g, which we prove first.

Lemma 2.2. Let f be continuous and strictly increasing on an open interval I, let $J = f(I)$, and let $g = f^{-1}\colon J \to I$.

(a) g is strictly increasing.

(b) Every $b \in J$ is an interior point of J; that is, J contains an open interval around b.

(c) g is continuous on J.

Proof. (a) Suppose $y_1 < y_2$ in J, but $g(y_1) \geq g(y_2)$. Applying the increasing function f gives $y_1 \geq y_2$, a contradiction.

(b) Let $a = g(b)$. Since I is open, there is an $h > 0$ with $[a-h, a+h] \subseteq I$. Then $f(a-h) < b < f(a+h)$, and by the IVT, f takes every value between $f(a-h)$ and $f(a+h)$ on $[a-h, a+h]$. So the open interval $\big(f(a-h), f(a+h)\big)$ lies in J and contains b.

(c) Let $b \in J$, let $a = g(b)$, and let $\varepsilon > 0$. Shrinking ε only makes what we must prove stronger, so we may assume that $[a-\varepsilon, a+\varepsilon] \subseteq I$. Then $f(a-\varepsilon) < b < f(a+\varepsilon)$, so

$$\delta = \min\big\{\, b - f(a-\varepsilon),\ f(a+\varepsilon) - b \,\big\} > 0.$$

If $y \in J$ and $|y-b| < \delta$, then $f(a-\varepsilon) < y < f(a+\varepsilon)$. Applying g, which is increasing by (a), gives $a - \varepsilon < g(y) < a + \varepsilon$; that is, $|g(y) - g(b)| < \varepsilon$. □

Proof of Theorem 2.1. We treat the case where f is increasing; the decreasing case is the same with the inequalities reversed. Since f is differentiable, it is continuous, so Lemma 2.2 applies. By part (b), g is defined on an open interval around b, so it makes sense to ask whether g is differentiable at b.

For $x \in I$ with $x \neq a$, let

$$Q(x) = \frac{f(x) - f(a)}{x - a}.$$

Then $Q(x) \to f'(a)$ as $x \to a$, and $Q(x) \neq 0$ because f is one-to-one. Now let $y \in J$ with $y \neq b$, and let $x = g(y)$. Then $f(x) = y$, and $x \neq a$ because g is one-to-one. So

$$\frac{g(y) - g(b)}{y - b} = \frac{x - a}{f(x) - f(a)} = \frac{1}{Q(x)} = \frac{1}{Q\big(g(y)\big)}.$$

Claim: $Q\big(g(y)\big) \to f'(a)$ as $y \to b$. Let $\varepsilon > 0$. Choose $\eta > 0$ such that $|Q(x) - f'(a)| < \varepsilon$ whenever $0 < |x-a| < \eta$. Since g is continuous at b (Lemma 2.2(c)), we can choose $\delta > 0$ such that $|g(y) - a| < \eta$ whenever $y \in J$ and $|y-b| < \delta$. If $0 < |y-b| < \delta$, then $g(y) \neq a$, so $0 < |g(y)-a| < \eta$, and hence $\big|Q\big(g(y)\big) - f'(a)\big| < \varepsilon$.

Finally, since $f'(a) \neq 0$, the quotient rule for limits gives

$$\lim_{y \to b} \frac{g(y) - g(b)}{y - b} = \lim_{y \to b} \frac{1}{Q\big(g(y)\big)} = \frac{1}{f'(a)}. \qquad \square$$

Remark. The heart of the proof is that $x = g(y) \to a$ as $y \to b$, which is the continuity of g. The hypothesis $f'(a) \neq 0$ is used only in the last step.

Corollary 2.3. If f is differentiable on an open interval I with $f' > 0$ on I (or $f' < 0$ on I), then f^{-1} is differentiable on $J = f(I)$, and

$$(f^{-1})'(x) = \frac{1}{f'\big(f^{-1}(x)\big)} \qquad \text{for all } x \in J.$$

Remark. The hypothesis $f'(a) \neq 0$ cannot be dropped. If $f'(a) = 0$ and g were differentiable at b, then the chain rule applied to $f\big(g(y)\big) = y$ at $y = b$ would give $f'(a)\, g'(b) = 1$, that is, $0 = 1$. So in this case g is *not* differentiable at b. For example, $f(x) = x^3$ is strictly increasing on $\mathbb{R}$ with $f'(0) = 0$, and its inverse $\sqrt[3]{x}$ is not differentiable at 0. Its graph has a vertical tangent there, which is the reflection of the horizontal tangent of x^3 at the origin.

Example 2.4. Return to $f(x) = x^3 + x$ from Example 1.4. Since $f'(x) = 3x^2 + 1 > 0$, Corollary 2.3 applies. To find $(f^{-1})'(2)$ we need $f^{-1}(2)$, which is 1 because $f(1) = 2$. So

$$(f^{-1})'(2) = \frac{1}{f'(1)} = \frac{1}{4},$$

and the tangent line to the graph of f^{-1} at $(2, 1)$ is $y = 1 + \frac{1}{4}(x - 2)$. Similarly, $f(2) = 10$ gives $(f^{-1})'(10) = 1/f'(2) = 1/13$.

The key step is always the same: to evaluate $(f^{-1})'$ at b, first find the a with $f(a) = b$.

Example 2.5. Since ln is differentiable on $(0, \infty)$ with derivative $1/x > 0$, Corollary 2.3 shows that e^x is differentiable on $\mathbb{R}$, and

$$\frac{d}{dx} e^x = \frac{1}{\ln'(e^x)} = \frac{1}{1/e^x} = e^x.$$

In particular, the corollary guarantees that e^x is differentiable, which is exactly what is needed to justify differentiating $\ln(e^x) = x$ with the chain rule.

Example 2.6. Let $f(x) = x^2$ on $(0, \infty)$, where $f'(x) = 2x > 0$. Its inverse is $\sqrt{x}$, so

$$\frac{d}{dx} \sqrt{x} = \frac{1}{f'(\sqrt{x})} = \frac{1}{2\sqrt{x}},$$

as expected. In the same way, for an integer $n \geq 2$, the inverse of x^n on $(0, \infty)$ is $x^{1/n}$, and

$$\frac{d}{dx} x^{1/n} = \frac{1}{n\big(x^{1/n}\big)^{n-1}} = \frac{1}{n}\, x^{\frac{1}{n} - 1}.$$

Example 2.7. Let $F(x) = \displaystyle\int_2^x \sqrt{t^4 + 9}\, dt$. Find $(F^{-1})'(0)$.

By the Fundamental Theorem of Calculus, $F'(x) = \sqrt{x^4 + 9} > 0$, so F is strictly increasing, and so one-to-one, on $\mathbb{R}$. Since $F(2) = \int_2^2 \sqrt{t^4 + 9}\, dt = 0$, we have $F^{-1}(0) = 2$. So

$$(F^{-1})'(0) = \frac{1}{F'(2)} = \frac{1}{\sqrt{25}} = \frac{1}{5}.$$

3 Inverse trigonometric functions

The trigonometric functions are periodic, so none of them is one-to-one. To invert one, we restrict it to a set where it is one-to-one and invert the restriction. The choice of set is a convention; we use the same choices as Salas.

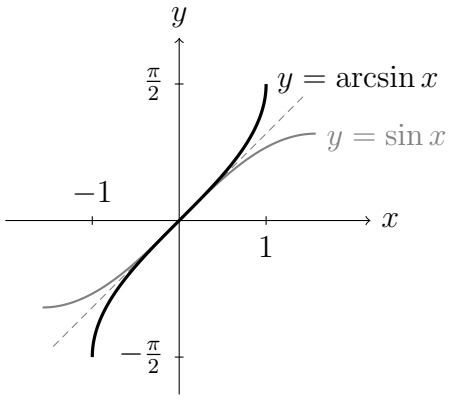
3.1 Arcsine

Since $\sin' = \cos > 0$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$, sin is strictly increasing on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and it goes from $\sin(-\frac{\pi}{2}) = -1$ to $\sin\frac{\pi}{2} = 1$. So on this interval it takes each value in $[-1, 1]$ exactly once: at least once by the IVT, and at most once because it is increasing.

Definition 3.1. For $-1 \le x \le 1$, $\arcsin x$ is the unique number θ in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ with $\sin\theta = x$.

In words: $\arcsin x$ is the angle between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ whose sine is x.

- The domain of arcsin is $[-1, 1]$, and its range is $[-\frac{\pi}{2}, \frac{\pi}{2}]$.
- $\sin(\arcsin x) = x$ for every $x \in [-1, 1]$.
- $\arcsin(\sin\theta) = \theta$ *only* for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.
- arcsin is increasing and odd: $\arcsin(-x) = -\arcsin x$.
- Notation: $\arcsin x$ is also written $\sin^{-1} x$. This does not mean $(\sin x)^{-1} = \csc x$, even though $\sin^2 x$ means $(\sin x)^2$.



Example 3.2. (a) $\arcsin\frac{1}{2} = \frac{\pi}{6}$, $\arcsin\big(-\frac{\sqrt{3}}{2}\big) = -\frac{\pi}{3}$, $\arcsin 1 = \frac{\pi}{2}$, and $\arcsin 0 = 0$.

(b) $\arcsin\big(\sin\frac{5\pi}{6}\big) = \arcsin\frac{1}{2} = \frac{\pi}{6}$, *not* $\frac{5\pi}{6}$, because $\frac{5\pi}{6}$ is not in $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

(c) $\arcsin(\sin 4) = \pi - 4$. Indeed, $\sin(\pi - 4) = \sin 4$, and since $\pi < 4 < \frac{3\pi}{2}$, the number $\pi - 4$ lies in $(-\frac{\pi}{2}, 0)$.

Theorem 3.3. arcsin is differentiable on $(-1, 1)$, and

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}} \qquad (-1 < x < 1).$$

Proof. Apply Corollary 2.3 to $f = \sin$ on the open interval $I = (-\frac{\pi}{2}, \frac{\pi}{2})$, where $f'(\theta) = \cos\theta > 0$; here $J = (-1, 1)$. For $-1 < x < 1$, let $\theta = \arcsin x$. Then

$$\frac{d}{dx} \arcsin x = \frac{1}{\cos\theta}.$$

Since $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $\cos\theta > 0$, so $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - x^2}$. □

Remark. For $0 < x < 1$ there is a picture: $\theta = \arcsin x$ is an angle in a right triangle with opposite side x and hypotenuse 1, so the adjacent side is $\sqrt{1 - x^2}$ and $\cos\theta = \sqrt{1 - x^2}$ (see Section 3.6). The triangle is a good mnemonic, but the sign argument in the proof is what handles negative x.

At $x = \pm 1$, arcsin is not differentiable: $\cos(\pm\frac{\pi}{2}) = 0$, so by the remark after Corollary 2.3, its graph has vertical tangents there.

By the chain rule, $\dfrac{d}{dx} \arcsin u = \dfrac{u'}{\sqrt{1 - u^2}}$.

Example 3.4. (a) $\dfrac{d}{dx} \arcsin(e^x) = \dfrac{e^x}{\sqrt{1 - e^{2x}}}$, for $x < 0$.

(b) $\dfrac{d}{dx}\left[x \arcsin x + \sqrt{1 - x^2}\right] = \arcsin x + \dfrac{x}{\sqrt{1 - x^2}} - \dfrac{x}{\sqrt{1 - x^2}} = \arcsin x$. So $\int \arcsin x \, dx = x \arcsin x + \sqrt{1 - x^2} + C$. In Section 4 we will see how to *find* this antiderivative.

3.2 Arctangent

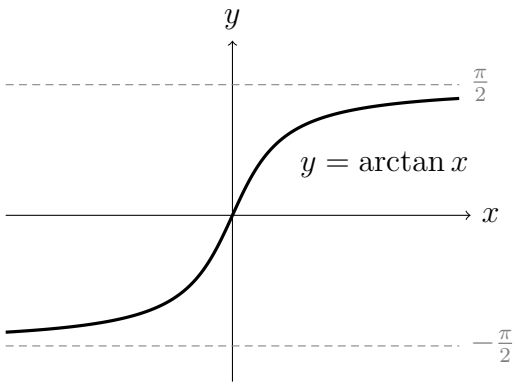
Since $\tan' = \sec^2 > 0$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$, tan is strictly increasing there. Also $\tan\theta \to \pm\infty$ as $\theta \to \pm\frac{\pi}{2}$, so by the IVT, tan takes every real value exactly once on $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Definition 3.5. For every real x, $\arctan x$ is the unique number θ in $(-\frac{\pi}{2}, \frac{\pi}{2})$ with $\tan\theta = x$.

- The domain of arctan is $\mathbb{R}$, and its range is $(-\frac{\pi}{2}, \frac{\pi}{2})$.
- $\tan(\arctan x) = x$ for every x, and $\arctan(\tan\theta) = \theta$ for $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$.
- arctan is increasing and odd, and

  $$\lim_{x\to\infty} \arctan x = \frac{\pi}{2}, \qquad \lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}.$$

  (Given $\theta < \frac{\pi}{2}$, every $x > \tan\theta$ has $\arctan x > \theta$.) So the graph has horizontal asymptotes $y = \pm\frac{\pi}{2}$.
- Values: $\arctan 1 = \frac{\pi}{4}$, $\arctan\frac{1}{\sqrt{3}} = \frac{\pi}{6}$, $\arctan(-\sqrt{3}) = -\frac{\pi}{3}$, $\arctan 0 = 0$.



Theorem 3.6. arctan is differentiable on $\mathbb{R}$, and

$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}.$$

Proof. Apply Corollary 2.3 to $f = \tan$ on $I = (-\frac{\pi}{2}, \frac{\pi}{2})$, where $f'(\theta) = \sec^2\theta > 0$; here $J = \mathbb{R}$. For any x, let $\theta = \arctan x$. Then

$$\frac{d}{dx}\arctan x = \frac{1}{\sec^2\theta} = \frac{1}{1+\tan^2\theta} = \frac{1}{1+x^2}.$$

(No sign issue arises this time.) □

Example 3.7. $\dfrac{d}{dx}\arctan\sqrt{x} = \dfrac{1}{1+x}\cdot\dfrac{1}{2\sqrt{x}} = \dfrac{1}{2\sqrt{x}\,(1+x)}$, for $x > 0$.

Proposition 3.8. For $x > 0$, $\arctan x + \arctan\dfrac{1}{x} = \dfrac{\pi}{2}$.

Proof. Let $g(x) = \arctan x + \arctan(1/x)$ for $x > 0$. Then

$$g'(x) = \frac{1}{1+x^2} + \frac{1}{1+1/x^2}\cdot\left(-\frac{1}{x^2}\right) = \frac{1}{1+x^2} - \frac{1}{x^2+1} = 0.$$

A function with zero derivative on an interval is constant (by the MVT), so g is constant on $(0,\infty)$, and $g(x) = g(1) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$. □

Remark. Geometrically: in a right triangle with legs 1 and x, the two acute angles are $\arctan x$ and $\arctan(1/x)$.

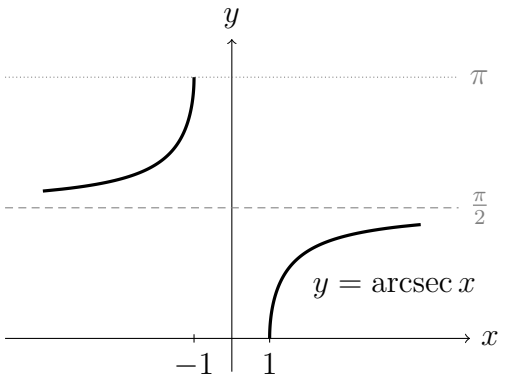
The same computation shows $g'(x) = 0$ for $x < 0$ too, but there $g(x) = -\frac{\pi}{2}$ (because arctan is odd). So g has derivative 0 on all of $\mathbb{R}\setminus\{0\}$ without being constant. This does not contradict the MVT, because $\mathbb{R}\setminus\{0\}$ is not an interval.

3.3 Arcsecant

On $[0,\pi]$, cos is strictly decreasing, so it is one-to-one. Removing $\frac{\pi}{2}$, where cos is 0, the function $\sec = 1/\cos$ is one-to-one on $[0,\frac{\pi}{2})\cup(\frac{\pi}{2},\pi]$. On $[0,\frac{\pi}{2})$ it takes every value in $[1,\infty)$, and on $(\frac{\pi}{2},\pi]$ every value in $(-\infty,-1]$.

Definition 3.9. For $|x| \geq 1$, $\operatorname{arcsec} x$ is the unique number θ in $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$ with $\sec\theta = x$; equivalently, with $\cos\theta = \frac{1}{x}$.

- The domain of arcsec is $(-\infty, -1] \cup [1, \infty)$, and its range is $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$.
- Values: $\operatorname{arcsec} 1 = 0$, $\operatorname{arcsec}\sqrt{2} = \frac{\pi}{4}$, $\operatorname{arcsec} 2 = \frac{\pi}{3}$, $\operatorname{arcsec}(-2) = \frac{2\pi}{3}$, $\operatorname{arcsec}(-\sqrt{2}) = \frac{3\pi}{4}$, $\operatorname{arcsec}(-1) = \pi$.
- $\operatorname{arcsec} x \to \frac{\pi}{2}$ as $x \to \pm\infty$, since $\cos(\operatorname{arcsec} x) = \frac{1}{x} \to 0$.
- arcsec is increasing on each of $[1, \infty)$ and $(-\infty, -1]$, but it is not increasing on its whole domain: $\operatorname{arcsec}(-2) > \operatorname{arcsec} 2$.



Theorem 3.10. arcsec is differentiable for $|x| > 1$, and

$$\frac{d}{dx}\operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}} \qquad (|x| > 1).$$

Proof. We have $\sec'\theta = \sec\theta\tan\theta = \dfrac{\sin\theta}{\cos^2\theta}$, which is positive on $(0, \frac{\pi}{2})$ and on $(\frac{\pi}{2}, \pi)$. Apply Corollary 2.3 separately on $I_1 = (0, \frac{\pi}{2})$, where $J_1 = (1, \infty)$, and on $I_2 = (\frac{\pi}{2}, \pi)$, where $J_2 = (-\infty, -1)$. For $|x| > 1$, let $\theta = \operatorname{arcsec} x$. Then

$$\frac{d}{dx}\operatorname{arcsec} x = \frac{1}{\sec\theta\tan\theta} = \frac{1}{x\tan\theta}.$$

Now $\tan^2\theta = \sec^2\theta - 1 = x^2 - 1$, so $\tan\theta = \pm\sqrt{x^2-1}$, and we need the sign:

- if $x > 1$, then $\theta \in (0, \frac{\pi}{2})$, so $\tan\theta > 0$;
- if $x < -1$, then $\theta \in (\frac{\pi}{2}, \pi)$, so $\tan\theta < 0$.

So $\tan\theta$ has the same sign as x, and $x\tan\theta = |x|\sqrt{x^2-1}$. □

Example 3.11. $\dfrac{d}{dx}\operatorname{arcsec}(x^2) = \dfrac{2x}{x^2\sqrt{x^4-1}} = \dfrac{2}{x\sqrt{x^4-1}}$, for $|x| > 1$. (Here $|x^2| = x^2$.)

Remark. Some textbooks restrict sec to $[0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$ instead. With that convention, $\frac{d}{dx}\operatorname{arcsec} x = \frac{1}{x\sqrt{x^2-1}}$ with no absolute value, but values such as $\operatorname{arcsec}(-2)$ change. Students who look at other references may run into this.

3.4 The other three

- $\arccos x$, for $-1 \le x \le 1$, is the unique $\theta \in [0, \pi]$ with $\cos\theta = x$. Since $\theta = \frac{\pi}{2} - \arcsin x$ lies in $[0, \pi]$ and $\cos\big(\frac{\pi}{2} - \arcsin x\big) = \sin(\arcsin x) = x$,

$$\arccos x = \frac{\pi}{2} - \arcsin x, \qquad \frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}.$$

- $\operatorname{arccot} x$, for all real x, is the unique $\theta \in (0, \pi)$ with $\cot\theta = x$. In the same way,

$$\operatorname{arccot} x = \frac{\pi}{2} - \arctan x, \qquad \frac{d}{dx}\operatorname{arccot} x = -\frac{1}{1+x^2}.$$

- $\operatorname{arccsc} x$, for $|x| \ge 1$, is the unique $\theta \in [-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$ with $\csc\theta = x$. Again,

$$\operatorname{arccsc} x = \frac{\pi}{2} - \operatorname{arcsec} x, \qquad \frac{d}{dx}\operatorname{arccsc} x = -\frac{1}{|x|\sqrt{x^2-1}}.$$

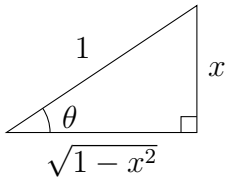
For integration these give nothing new, since each derivative is the negative of one we already have.

3.5 Summary

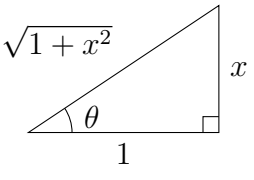
Function	**Domain**	**Range**	**Derivative**
$\arcsin x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	$\dfrac{1}{\sqrt{1-x^2}} \quad (\lvert x\rvert < 1)$
$\arctan x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	$\dfrac{1}{1+x^2}$
$\operatorname{arcsec} x$	$\lvert x\rvert \ge 1$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$	$\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}} \quad (\lvert x\rvert > 1)$
$\arccos x$	$[-1, 1]$	$[0, \pi]$	$-\dfrac{1}{\sqrt{1-x^2}} \quad (\lvert x\rvert < 1)$
$\operatorname{arccot} x$	$\mathbb{R}$	$(0, \pi)$	$-\dfrac{1}{1+x^2}$
$\operatorname{arccsc} x$	$\lvert x\rvert \ge 1$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$	$-\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}} \quad (\lvert x\rvert > 1)$

3.6 Simplifying compositions

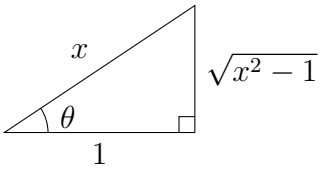
Expressions like $\tan(\arcsin x)$ can be rewritten without trigonometric functions. The method: name the angle, use a Pythagorean identity, and use the range of the inverse function to fix the sign. A right triangle (drawn for $x > 0$) shows what the answer should be.



$\theta = \arcsin x$



$\theta = \arctan x$



$\theta = \operatorname{arcsec} x$

Example 3.12. Simplify $\tan(\arcsin x)$ for $-1 < x < 1$.

Let $\theta = \arcsin x$, so $\sin\theta = x$. Since $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\cos\theta \geq 0$, so $\cos\theta = \sqrt{1-x^2}$. Therefore $\tan(\arcsin x) = \dfrac{x}{\sqrt{1-x^2}}$.

Example 3.13. Simplify $\cos(\arctan x)$ and $\sin(\arctan x)$.

Let $\theta = \arctan x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Then $\cos\theta > 0$ and $\sec^2\theta = 1 + \tan^2\theta = 1 + x^2$, so

$$\cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}, \qquad \sin(\arctan x) = \tan\theta\cos\theta = \frac{x}{\sqrt{1+x^2}}.$$

For instance, $\sin(2\arctan x) = 2\sin\theta\cos\theta = \dfrac{2x}{1+x^2}$.

Example 3.14. Find $\sin\big(\operatorname{arcsec}(-3)\big)$.

Let $\theta = \operatorname{arcsec}(-3)$, so $\cos\theta = -\frac{1}{3}$ and $\theta \in (\frac{\pi}{2}, \pi)$. There $\sin\theta > 0$, so $\sin\theta = \sqrt{1 - \frac{1}{9}} = \frac{2\sqrt{2}}{3}$. (The triangle alone would not tell us the sign.)

When we get to trigonometric substitution, we will use these same triangles in reverse.

3.7 Integrals

Reading the derivative formulas backwards gives

$$\boxed{\begin{aligned} \int \frac{dx}{\sqrt{1-x^2}} &= \arcsin x + C, \\ \int \frac{dx}{1+x^2} &= \arctan x + C, \\ \int \frac{dx}{x\sqrt{x^2-1}} &= \operatorname{arcsec}|x| + C. \end{aligned}}$$

The third formula needs a comment. For $x > 1$ it is the derivative formula, since $|x| = x$. For $x < -1$,

$$\frac{d}{dx}\operatorname{arcsec}|x| = \frac{d}{dx}\operatorname{arcsec}(-x) = -\frac{1}{|x|\sqrt{x^2-1}} = \frac{1}{x\sqrt{x^2-1}},$$

since $-|x| = x$ when $x < 0$. Compare $\int \frac{dx}{x} = \ln|x| + C$.

More generally, for a constant $a > 0$,

$$\boxed{\begin{aligned} \int \frac{dx}{\sqrt{a^2 - x^2}} &= \arcsin \frac{x}{a} + C, \\ \int \frac{dx}{a^2 + x^2} &= \frac{1}{a} \arctan \frac{x}{a} + C, \\ \int \frac{dx}{x\sqrt{x^2 - a^2}} &= \frac{1}{a} \operatorname{arcsec} \frac{|x|}{a} + C. \end{aligned}}$$

These follow from the first three by the substitution $u = x/a$. For example, with $x = au$ and $dx = a\,du$,

$$\int \frac{dx}{a^2 + x^2} = \int \frac{a\,du}{a^2(1 + u^2)} = \frac{1}{a} \arctan u + C = \frac{1}{a} \arctan \frac{x}{a} + C.$$

Rather than memorizing the general versions, one can always substitute.

Example 3.15. (a) $\displaystyle\int_0^1 \frac{dx}{1 + x^2} = \arctan 1 - \arctan 0 = \frac{\pi}{4}$. (So Riemann sums for $\frac{4}{1+x^2}$ on $[0, 1]$ approximate π.)

(b) $\displaystyle\int_0^{1/2} \frac{dx}{\sqrt{1 - x^2}} = \arcsin \frac{1}{2} - \arcsin 0 = \frac{\pi}{6}$.

Example 3.16. $\displaystyle\int \frac{dx}{\sqrt{4 - 9x^2}}$. Let $u = 3x$, so $du = 3\,dx$:

$$\int \frac{dx}{\sqrt{4 - 9x^2}} = \frac{1}{3} \int \frac{du}{\sqrt{4 - u^2}} = \frac{1}{3} \arcsin \frac{u}{2} + C = \frac{1}{3} \arcsin \frac{3x}{2} + C.$$

Example 3.17. Compare $\displaystyle\int \frac{dx}{1 + x^2} = \arctan x + C$ with $\displaystyle\int \frac{x\,dx}{1 + x^2}$, where $u = 1 + x^2$ gives $\frac{1}{2} \ln(1 + x^2) + C$. Similar-looking integrands can have very different antiderivatives. Splitting a fraction uses both:

$$\int \frac{x + 3}{1 + x^2}\,dx = \int \frac{x}{1 + x^2}\,dx + 3 \int \frac{dx}{1 + x^2} = \frac{1}{2} \ln(1 + x^2) + 3 \arctan x + C.$$

Example 3.18. $\displaystyle\int \frac{x}{1 + x^4}\,dx$. Let $u = x^2$, so $du = 2x\,dx$:

$$\int \frac{x}{1 + x^4}\,dx = \frac{1}{2} \int \frac{du}{1 + u^2} = \frac{1}{2} \arctan u + C = \frac{1}{2} \arctan(x^2) + C.$$

Example 3.19 (Completing the square)**.** (a) $\displaystyle\int \frac{dx}{x^2 + 4x + 13}$. Since $x^2 + 4x + 13 = (x + 2)^2 + 9$, let $u = x + 2$:

$$\int \frac{dx}{(x + 2)^2 + 9} = \int \frac{du}{u^2 + 9} = \frac{1}{3} \arctan \frac{u}{3} + C = \frac{1}{3} \arctan \frac{x + 2}{3} + C.$$

(b) $\displaystyle\int \frac{dx}{\sqrt{6x - x^2}}$. Since $6x - x^2 = 9 - (x - 3)^2$, let $u = x - 3$:

$$\int \frac{dx}{\sqrt{9 - (x - 3)^2}} = \int \frac{du}{\sqrt{9 - u^2}} = \arcsin \frac{u}{3} + C = \arcsin \frac{x - 3}{3} + C.$$

Example 3.20. $\displaystyle\int \frac{e^x}{\sqrt{1-e^{2x}}}\,dx$. Let $u = e^x$, so $du = e^x\,dx$:

$$\int \frac{e^x}{\sqrt{1-e^{2x}}}\,dx = \int \frac{du}{\sqrt{1-u^2}} = \arcsin(e^x) + C.$$

Example 3.21. (a) $\displaystyle\int_{\sqrt{2}}^{2} \frac{dx}{x\sqrt{x^2-1}} = \operatorname{arcsec} 2 - \operatorname{arcsec}\sqrt{2} = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}.$

(b) $\displaystyle\int_{-2}^{-\sqrt{2}} \frac{dx}{x\sqrt{x^2-1}} = \Big[\operatorname{arcsec}|x|\Big]_{-2}^{-\sqrt{2}} = \operatorname{arcsec}\sqrt{2} - \operatorname{arcsec} 2 = -\frac{\pi}{12}$. This should be negative, since the integrand is negative for $x < 0$. Dropping the absolute value would give $\operatorname{arcsec}(-\sqrt{2}) - \operatorname{arcsec}(-2) = \frac{3\pi}{4} - \frac{2\pi}{3} = \frac{\pi}{12}$, which has the wrong sign.

Example 3.22. For $b > 0$, the area under $y = \frac{1}{1+x^2}$ from 0 to b is

$$A(b) = \int_0^b \frac{dx}{1+x^2} = \arctan b,$$

and $A(b) \to \frac{\pi}{2}$ as $b \to \infty$. So although the region under the whole curve is infinitely long, the area of the part with $0 \le x \le b$ never exceeds $\frac{\pi}{2}$. By symmetry, the same holds on the left, for a total of π.

4 Integration by parts

Substitution is the chain rule run backwards; integration by parts is the product rule run backwards.

Theorem 4.1 (Integration by parts)**.** Suppose u and v have continuous derivatives on an interval. Then

$$\int u(x)\,v'(x)\,dx = u(x)\,v(x) - \int v(x)\,u'(x)\,dx,$$

and for a and b in the interval,

$$\int_a^b u(x)\,v'(x)\,dx = \Big[u(x)\,v(x)\Big]_a^b - \int_a^b v(x)\,u'(x)\,dx.$$

Proof. By the product rule, the derivative of the right side of the first formula is $(u'v + uv') - vu' = uv'$, so the right side is an antiderivative of uv'. For the second formula, integrate $(uv)' = u'v + uv'$ from a to b. By the Fundamental Theorem of Calculus, $\int_a^b (uv)'\,dx = \big[uv\big]_a^b$. (All the integrands are continuous, so all the integrals exist.) □

Shorthand. With $du = u'(x)\,dx$ and $dv = v'(x)\,dx$, the formula reads

$$\boxed{\int u\,dv = uv - \int v\,du.}$$

How to choose. Write the integrand as $u \cdot dv$, where

- you can integrate dv to find v, and
- $\int v\,du$ is simpler than the original integral.

Rule of thumb: take u to be something that becomes simpler when differentiated (a polynomial, $\ln x$, an inverse trigonometric function), and dv to be something that does not become worse when integrated (e^x, $\sin x$, $\cos x$, or just dx).

Any antiderivative v works, so we always take the simplest one, with constant 0. (Replacing v by $v + c$ adds cu to uv and cu to $\int v\,du$, and these cancel.)

Example 4.2. $\int xe^x\,dx$. Let $u = x$ and $dv = e^x\,dx$, so $du = dx$ and $v = e^x$:

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C.$$

Check: $\frac{d}{dx}(xe^x - e^x) = e^x + xe^x - e^x = xe^x$.

A bad choice: $u = e^x$ and $dv = x\,dx$ give $du = e^x\,dx$ and $v = \frac{x^2}{2}$, so

$$\int xe^x\,dx = \frac{x^2}{2}e^x - \int \frac{x^2}{2}e^x\,dx,$$

which is correct but leaves a harder integral than the one we started with.

Example 4.3. $\int_0^\pi x\sin x\,dx$. Let $u = x$ and $dv = \sin x\,dx$, so $du = dx$ and $v = -\cos x$:

$$\int_0^\pi x\sin x\,dx = \Big[-x\cos x\Big]_0^\pi + \int_0^\pi \cos x\,dx = \pi + \Big[\sin x\Big]_0^\pi = \pi.$$

Example 4.4. $\int \ln x\,dx$. There is no visible product, but we can take $u = \ln x$ and $dv = dx$, so $du = \frac{dx}{x}$ and $v = x$:

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - x + C.$$

Example 4.5. $\int x\ln x\,dx$. Here the rule of thumb says to take $u = \ln x$, even though x is a polynomial: we can't easily integrate $\ln x$, but differentiating it makes it simpler. With $u = \ln x$ and $dv = x\,dx$, we have $du = \frac{dx}{x}$ and $v = \frac{x^2}{2}$, so

$$\int x\ln x\,dx = \frac{x^2}{2}\ln x - \int \frac{x^2}{2}\cdot\frac{1}{x}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C.$$

Example 4.6 (Inverse trigonometric functions)**.** (a) $\int \arctan x\,dx$. Let $u = \arctan x$ and $dv = dx$, so $du = \frac{dx}{1+x^2}$ and $v = x$:

$$\int \arctan x\,dx = x\arctan x - \int \frac{x}{1+x^2}\,dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + C.$$

(b) $\int \arcsin x\,dx$. Let $u = \arcsin x$ and $dv = dx$, so $du = \frac{dx}{\sqrt{1-x^2}}$ and $v = x$:

$$\int \arcsin x\,dx = x\arcsin x - \int \frac{x}{\sqrt{1-x^2}}\,dx.$$

For the last integral, let $w = 1-x^2$, so $dw = -2x\,dx$: $\int \frac{x\,dx}{\sqrt{1-x^2}} = -\frac{1}{2}\int w^{-1/2}\,dw = -\sqrt{1-x^2}+C$. So

$$\int \arcsin x\,dx = x\arcsin x + \sqrt{1-x^2} + C,$$

which agrees with what we checked in Section 3.

Example 4.7 (Integrating by parts twice)**.** $\int x^2e^x\,dx$. Let $u = x^2$ and $dv = e^x\,dx$, so $du = 2x\,dx$ and $v = e^x$:

$$\int x^2e^x\,dx = x^2e^x - 2\int xe^x\,dx = x^2e^x - 2(xe^x - e^x) + C = (x^2-2x+2)e^x + C,$$

using the earlier example for $\int xe^x\,dx$.

The same choice for $\int x^ne^x\,dx$ gives the *reduction formula*

$$\int x^ne^x\,dx = x^ne^x - n\int x^{n-1}e^x\,dx,$$

which lowers the power of x by one each time. For example,

$$\int x^3e^x\,dx = x^3e^x - 3\int x^2e^x\,dx = (x^3-3x^2+6x-6)e^x + C.$$

Example 4.8 (Solving for the integral)**.** $I = \int e^x\sin x\,dx$. Let $u = e^x$ and $dv = \sin x\,dx$, so $du = e^x\,dx$ and $v = -\cos x$:

$$I = -e^x\cos x + \int e^x\cos x\,dx.$$

Integrate by parts again with $u = e^x$ and $dv = \cos x\,dx$, so $v = \sin x$:

$$\int e^x\cos x\,dx = e^x\sin x - \int e^x\sin x\,dx = e^x\sin x - I.$$

So $I = -e^x\cos x + e^x\sin x - I$, and solving for I,

$$\int e^x\sin x\,dx = \frac{1}{2}e^x(\sin x - \cos x) + C.$$

Check: $\frac{d}{dx}\left[\frac{1}{2}e^x(\sin x - \cos x)\right] = \frac{1}{2}e^x(\sin x - \cos x) + \frac{1}{2}e^x(\cos x + \sin x) = e^x\sin x$.

Warning: in the second step we must again take $u = e^x$. Choosing $u = \cos x$ and $dv = e^x\,dx$ instead just undoes the first step and leads to $I = I$.

Example 4.9 (Substitution and parts together)**.** $\int x^3e^{x^2}\,dx$. Write the integrand as $x^2\cdot xe^{x^2}$, and let $u = x^2$ and $dv = xe^{x^2}\,dx$. Then $du = 2x\,dx$ and, by the substitution $w = x^2$, $v = \frac{1}{2}e^{x^2}$. So

$$\int x^3e^{x^2}\,dx = \frac{1}{2}x^2e^{x^2} - \int xe^{x^2}\,dx = \frac{1}{2}x^2e^{x^2} - \frac{1}{2}e^{x^2} + C = \frac{1}{2}(x^2-1)e^{x^2} + C.$$

(Alternatively, substitute $w = x^2$ first to get $\frac{1}{2}\int we^w\,dw$, and then integrate by parts.)

Example 4.10 (A volume)**.** The region under $y = \cos x$ for $0 \le x \le \frac{\pi}{2}$ is revolved about the y-axis. By the shell method, the volume is

$$V = \int_0^{\pi/2} 2\pi x \cos x \, dx.$$

Let $u = x$ and $dv = \cos x \, dx$, so $du = dx$ and $v = \sin x$:

$$\int_0^{\pi/2} x \cos x \, dx = \Big[x \sin x\Big]_0^{\pi/2} - \int_0^{\pi/2} \sin x \, dx = \frac{\pi}{2} - \Big[-\cos x\Big]_0^{\pi/2} = \frac{\pi}{2} - 1.$$

So $V = 2\pi\big(\frac{\pi}{2} - 1\big) = \pi^2 - 2\pi$.

Optional: integrating an inverse function. The integrals of $\ln x$, $\arctan x$, and $\arcsin x$ above all have the same shape, and the reason ties together everything in these notes.

Proposition 4.11. Let f be differentiable with $f' > 0$ (or $f' < 0$) on an open interval, and let F be an antiderivative of f. Then

$$\int f^{-1}(x) \, dx = x \, f^{-1}(x) - F\big(f^{-1}(x)\big) + C.$$

Proof. By Corollary 2.3, f^{-1} is differentiable, so we can differentiate the right side:

$$f^{-1}(x) + x \, (f^{-1})'(x) - f\big(f^{-1}(x)\big) \, (f^{-1})'(x) = f^{-1}(x),$$

since $f\big(f^{-1}(x)\big) = x$. □

This is what integration by parts with $u = f^{-1}(x)$ and $dv = dx$ produces, followed by the substitution $y = f^{-1}(x)$ in $\int x \, (f^{-1})'(x) \, dx$. Examples:

- $f = \exp$, $F = \exp$: $\int \ln x \, dx = x \ln x - e^{\ln x} + C = x \ln x - x + C$.
- $f = \sin$, $F = -\cos$: $\int \arcsin x \, dx = x \arcsin x + \cos(\arcsin x) + C = x \arcsin x + \sqrt{1 - x^2} + C$.
- $f = \tan$, $F(y) = \ln|\sec y|$: $\int \arctan x \, dx = x \arctan x - \ln\big|\sec(\arctan x)\big| + C = x \arctan x - \frac{1}{2}\ln(1 + x^2) + C$, since $\sec(\arctan x) = \sqrt{1 + x^2}$.